Exercise 3: The numerical toolbox

Prognostic Modelling (pea700), WiSe 2026/27

Out: week 3 · Back: week 4 · 13 points

Covers unit 3 of the script. Problem 1 is worked on in the lab; problems 2 to 4 are homework.

Your toolbox. From this sheet on, collect the functions you write that are useful beyond one problem, such as the quadrature rules, bisection and Newton's method, in a file `toolbox.py` in your repository. In a later sheet, or in your project, you then write `from toolbox import bisection` and use your own, tested function instead of writing it again.

Problem 1: Truncation and round-off error (in the lab, 4 points)

Approximate the derivative of $u(x) = e^{-x^2}$ at $x = 0.5$ with the forward and the central difference.

a) Plot the absolute error of both as a function of h for $h = 10^{-1}, \dots, 10^{-14}$ with logarithmic axes.

b) Determine the observed order of both formulas in a range of h where round-off plays no role.

c) Read the optimal step and the smallest error from your plot and compare them with the estimates $h \sim \sqrt{\varepsilon}$ and $h \sim \varepsilon^{1/3}$ from the script.

Problem 2: A one-sided difference (4 points)

At a boundary at x only the points x, $x + h$ and $x + 2h$ are available.

a) Use the Taylor series to find α, β, γ such that $[\alpha\, u(x) + \beta\, u(x + h) + \gamma\, u(x + 2h)]\,/h$ approximates $u'(x)$ with the highest possible order. Give the leading term of the truncation error.

b) Confirm the order and the leading error term with a convergence test for a function of your choice.

Problem 3: Quadrature (2 points)

Implement the composite trapezoidal rule and Simpson's rule. Compute $\int_0^\pi \sin x \, \mathrm{d}x = 2$ with $N = 4, 8, 16, 32$ subintervals and determine the observed order of both rules.

Problem 4: Root finding (3 points)

How deep is the water in a river that carries a given discharge? For a channel with a rectangular cross section of width b, Manning's formula gives the discharge at the water depth y:

$$Q(y) = \frac{\sqrt{S}}{n} A R^{2/3}, \qquad A = b\,y, \qquad R = \frac{A}{b + 2y}.$$

A is the area of the cross section, and the hydraulic radius R is this area divided by the wetted perimeter $b + 2y$ (bottom and both banks). Take the width $b = 10$ m, the bed slope $S = 0.001$ and the roughness $n = 0.035$ s m$^{-1/3}$ of a natural river bed. We seek the depth at which the river carries $Q_0 = 50$ m^3 s^{-1}, the root of $f(y) = Q(y) - Q_0 = 0$.

a) Write two functions of your own, `bisection(f, a, b, tol)` and `newton(f, dfdx, x0, tol)`. Each returns the root and the number of iterations it needed.

b) The derivative is

$$f'(y) = Q(y) \left(\frac{5}{3y} - \frac{4}{3(b + 2y)} \right).$$

Check it with a central difference before you use it. Find the depth with bisection and with Newton's method to a tolerance of 10^{-12} m. Justify your bracket and starting value with a plot. Compare the number of iterations.

c) Start Newton's method at $y_0 = 0.01$ m. What happens, and why? Why can it not start at $y_0 = 0$?